5. Estimation of oxalic acid

Aim :

To estimate the amount of oxalic acid dissolved in 1250 ml of the given unknown solution volumetrically. For this you are given with a standard solution of HCl solution of normality 0.1010 N and sodium hydroxide solution as link solution.

Principle:

Neutralization of Sodium hydroxide by HCl is given below. To indicate the end point, phenolphthalein is used as an indicator.

$$NaOH + HCl \longrightarrow NaCl + H_2O$$

Neutralization of Sodium hydroxide by oxalic acid is given below. To indicate the end point, phenolphthalein is used as an indicator.

$$2NaOH + \underset{\text{Oxalic acid}}{(COOH)_2} \longrightarrow \underset{\text{Sodium oxalate}}{(COONa)_2} + 2H_2O$$

Short procedure:

s.no	Content	Titration-I	Titration-II
1	Burette solution	HCl (standard solution)	Oxalic acid (unknown solution)
2	Pipette solution	20 ml of NaOH link solution	20 ml of NaOH link solution
4	Temperature	Lab temperature	Lab temperature
5	Indicator	Phenolphthalein	Phenolphthalein
6	End point	Disappearance of pink colour	Disappearance of pink colour
7	Equivalent weight of oxalic acid = 63		

Procedure :

Titration–I

(standard HCl)Vs (link NaOH)

Burette is washed with water, rinsed with HCl solution and filled with same HCl solution up to the zero mark. Exactly 20 ml of NaOH is pipetted out into the clean, washed conical flask. To This solution 2 to 3 drops of phenolphthalein indicator is added and titrated against HCl solution from the burette. HCl is added drop wise till the pink colour disappears completely. Burette reading is noted and the same procedure is repeated to get concordant values.

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